

GDP Growth & Economic Performance:

A Comparative Analysis for Banking Sector Decision-Making

AnalystLab Africa — Batch B Internship

Week 8 Capstone Project

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Data Source: World Bank World Development Indicators (WDI)

1. Objective / Problem Statement

This project applies the end-to-end data analytics workflow to World Bank World Development Indicators (WDI) data in order to evaluate the economic growth and stability profile of Nigeria relative to a set of comparator economies. The analysis is framed specifically around questions a banking sector stakeholder would ask when assessing market opportunity, lending risk, and investment strategy in Nigeria: How fast is the economy growing? How does growth translate into per-capita prosperity? What inflation, unemployment, and foreign investment risks accompany that growth? And how does Nigeria compare to peer emerging markets on each of these dimensions?

2. Dataset Description

The dataset used is the World Bank's World Development Indicators (WDI), a global collection of development statistics covering over 200 countries and territories from 1960 to the present. The full raw file (WDICSV.csv) contains over 1,400 indicators across themes including economy, poverty, education, health, environment, and governance.

For this analysis, the dataset was filtered to 10 countries and 9 indicators relevant to a GDP growth and economic performance theme, covering the years 2000–2025:

Countries analyzed: Nigeria, South Africa, Egypt, Kenya, Ghana, India, Indonesia, Brazil, United States, China

Indicator	WDI Code
GDP (current US\$)	NY.GDP.MKTP.CD
GDP growth (annual %)	NY.GDP.MKTP.KD.ZG
GDP per capita (current US\$)	NY.GDP.PCAP.CD
Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG
Foreign direct investment, net inflows (% of GDP)	BX.KLT.DINV.WD.GD.ZS
Gross capital formation (% of GDP)	NE.GDI.TOTL.ZS
Exports of goods and services (% of GDP)	NE.EXP.GNFS.ZS
Imports of goods and services (% of GDP)	NE.IMP.GNFS.ZS
Unemployment, total (% of labor force)	SL.UEM.TOTL.ZS

3. Data Cleaning Process

- Imported the raw WDICSV.csv (198MB, 400,000+ rows) into Microsoft SQL Server Management Studio (SSMS) using the Import Flat File wizard, alongside supporting metadata files WDIcountry.csv and WDIseries.csv.

- Resolved column truncation errors during import by widening text fields (e.g., Country Name, Indicator Name, and several free-text description fields) to accommodate long entries such as regional aggregate names and detailed indicator definitions.
- Removed a phantom header row that had been imported as a literal data row (containing the text "Country Name", "Country Code", etc.) rather than being used as column headers.
- Renamed all 70 auto-generated columns (column1–column70) into meaningful names: CountryName, CountryCode, IndicatorName, IndicatorCode, and Y1960 through Y2025 for each year column, using a T-SQL loop to automate the renaming and avoid manual error.
- Filtered the full dataset down to a working table (WDI_GDP_Filtered) containing only the 10 selected countries and 9 GDP-related indicators, explicitly excluding aggregate regional groupings (e.g., "Africa Eastern and Southern") to avoid double-counting in comparisons.
- Resolved a data type mismatch across year columns (one column had been auto-typed differently from the rest) by explicitly casting all year columns to FLOAT before reshaping.
- Reshaped the data from wide format (one column per year) to long format (CountryName, CountryCode, IndicatorName, IndicatorCode, Year, Value) using SQL's UNPIVOT operator, restricted to the 2000–2025 range. This step also automatically excluded NULL values, meaning the resulting table contains only years and indicators with actual reported data.
- Validated the cleaned table (WDI_GDP_Long, 2,253 rows) by spot-checking Nigeria's GDP growth trend against known historical events — confirming a -1.62% contraction in 2016 (oil price crash recession) and a -6.37% contraction in 2020 (COVID-19), both consistent with documented economic history.

4. Analysis Methodology

With a clean, validated long-format table in place, the analysis proceeded in three stages, each built directly into its own Power BI dashboard page:

- **Overview (Page 1):** Established baseline GDP fundamentals across all 10 countries — total GDP, GDP growth rate, and GDP per capita — to distinguish economic scale from per-person prosperity.
- **Comparative Deep Dive (Page 2):** Directly compared Nigeria against South Africa (used as a key African peer benchmark) across inflation, GDP growth, and foreign direct investment (FDI), surfacing the trade-off between growth rate and price stability.
- **Banking Risk Indicators (Page 3):** Focused specifically on the three metrics most relevant to a bank's risk assessment — inflation, unemployment, and FDI — both as latest-year snapshots and as 26-year trends.

DAX measures were created in Power BI to isolate each indicator for the latest reported year per country (e.g., Nigeria GDP, Nigeria Inflation, Nigeria Unemployment, Nigeria FDI, South Africa GDP Growth, South Africa Inflation), and interactive slicers (Country, Year, Indicator) were added to each page to allow dynamic exploration of the underlying data.

5. Key Findings

- Nigeria's GDP reached approximately \$290.79 billion, with GDP growth of 4.01% in the most recent reported year — outpacing South Africa's 1.11% growth over the same period.
- Despite stronger aggregate growth, Nigeria's GDP per capita (\$1,224.25) remains among the lowest of the 10 countries analyzed, far below the United States (~\$85,000–90,000) and China, reflecting the effect of Nigeria's large population on a per-person basis.
- Nigeria's inflation rate (23.01%) is roughly seven times South Africa's (3.21%), and shows a sharp upward trajectory from 2021 onward — consistent with the effects of subsidy removal and currency devaluation during that period.
- Nigeria's FDI net inflows (1.38% of GDP) sit mid-pack among comparator countries — behind Egypt and Brazil, but ahead of South Africa, which recorded net FDI outflows in the same year.
- Nigeria's official unemployment rate (3.06%) is far lower than South Africa's (over 32%), though this headline figure likely understates real labor market slack given the size of Nigeria's informal sector.

6. Insights

Growth vs. prosperity: Nigeria's strong headline GDP growth does not yet translate into comparable per-capita prosperity, suggesting that population growth is currently outpacing per-person income gains — a key consideration for consumer-facing banking products versus large-scale corporate or infrastructure financing.

Growth-inflation trade-off: Nigeria's combination of above-peer growth and substantially elevated inflation points to an economy expanding rapidly but carrying meaningful currency and pricing risk, which has direct implications for loan pricing and foreign-currency exposure.

Investment and informality: Modest FDI inflows alongside a low official unemployment rate suggest a market where growth may be driven substantially by informal-sector activity rather than large-scale formal investment — a nuance that headline statistics alone can obscure.

7. Recommendations

- Banks and lenders operating in or entering the Nigerian market should factor elevated inflation into loan pricing and consider inflation-indexed or shorter-duration lending products to manage currency devaluation risk.
- Given the gap between GDP growth and GDP per capita, retail and consumer banking strategies should be paired with financial inclusion initiatives that reach a broader base of the population, rather than assuming rising GDP alone signals rising consumer purchasing power.
- Investment and corporate banking teams should treat Nigeria's FDI position as a signal of cautious but real investor interest, and could explore products or advisory services that de-risk foreign investment entry (e.g., hedging instruments, political risk insurance partnerships).
- Given the likely underreporting of informal-sector unemployment, banks assessing SME and micro-lending opportunities should supplement official unemployment statistics with informal-economy indicators before finalizing risk models.
- Continued monitoring of Nigeria relative to peer economies (South Africa, Egypt, Kenya, Ghana) is recommended, as relative positioning on inflation and FDI can shift meaningfully year to year.

8. Conclusion

This project applied a complete data analytics workflow — from raw World Bank data through SQL-based cleaning and transformation to an interactive three-page Power BI dashboard — to build a banking-sector-relevant view of Nigeria's economic performance. The analysis shows an economy with genuine growth momentum, but one where inflation, per-capita prosperity, and informal-sector activity each add important nuance that headline growth figures alone do not capture. These findings demonstrate how macroeconomic indicator analysis can directly inform practical banking and investment decision-making in an emerging market context.